Ratio and Proportion – Formula Sheet

◆ 1. Ratio

- A ratio compares two quantities
  [
  Ratio = \frac{a}{b} = a:b
  ]
- If (a:b = c:d)
  then (\frac{a}{b} = \frac{c}{d})

◆ 2. Proportion

- When two ratios are equal:
  [
  a:b = c:d
  ]
- Cross multiplication:
  [
  a \times d = b \times c
  ]

◆ 3. Continued Proportion

If:
[
a:b = b:c
]

Then:
[
b^2 = a \times c
]

◆ 4. Mean Proportional

[
\text{Mean Proportional} = \sqrt{a \times b}
]

◆ 5. Third Proportional

If:
[
a:b = b:c
]

Then:
[
c = \frac{b^2}{a}
]

◆ 6. Fourth Proportional

If:
[
a:b = c:d
]

Then:
[
d = \frac{b \times c}{a}
]

◆ 7. Duplicate Ratio

[
a^2 : b^2
]

◆ 8. Sub-duplicate Ratio

[
\sqrt{a} : \sqrt{b}
]

◆ 9. Compound Ratio

[
(a:b) \times (c:d) = (a \times c):(b \times d)
]

◆ 10. Inverse Ratio

[
a:b = \frac{1}{b} : \frac{1}{a}
]

◆ 11. Componendo & Dividendo

If:
[
\frac{a}{b} = \frac{c}{d}
]

Then:
[
\frac{a+b}{a-b} = \frac{c+d}{c-d}
]

◆ 12. Variation

Direct Variation:

[
x \propto y \Rightarrow x = ky
]

Inverse Variation:

[
x \propto \frac{1}{y} \Rightarrow x = \frac{k}{y}
]

◆ 13. Important Shortcuts

- If:
  [
  a:b = m:n

]
Then:
[
a = km,\quad b = kn
]

- Sum method:
  If total = T
  [
  a = \frac{m}{m+n} \times T,\quad b = \frac{n}{m+n} \times T
  ]

◆ 14. Ratio Change Formula

If a quantity changes from (a) to (b):
[
\text{Ratio} = a:b
]

◆ 15. Partnership Formula

Profit sharing:
[
\text{Profit} \propto (\text{Investment} \times \text{Time})
]

◆ 16. Mixture Formula

If two ingredients:
[
\text{Ratio} = \frac{\text{Difference of opposite values}}{\text{Difference of other side}}
]

(Alligation method)